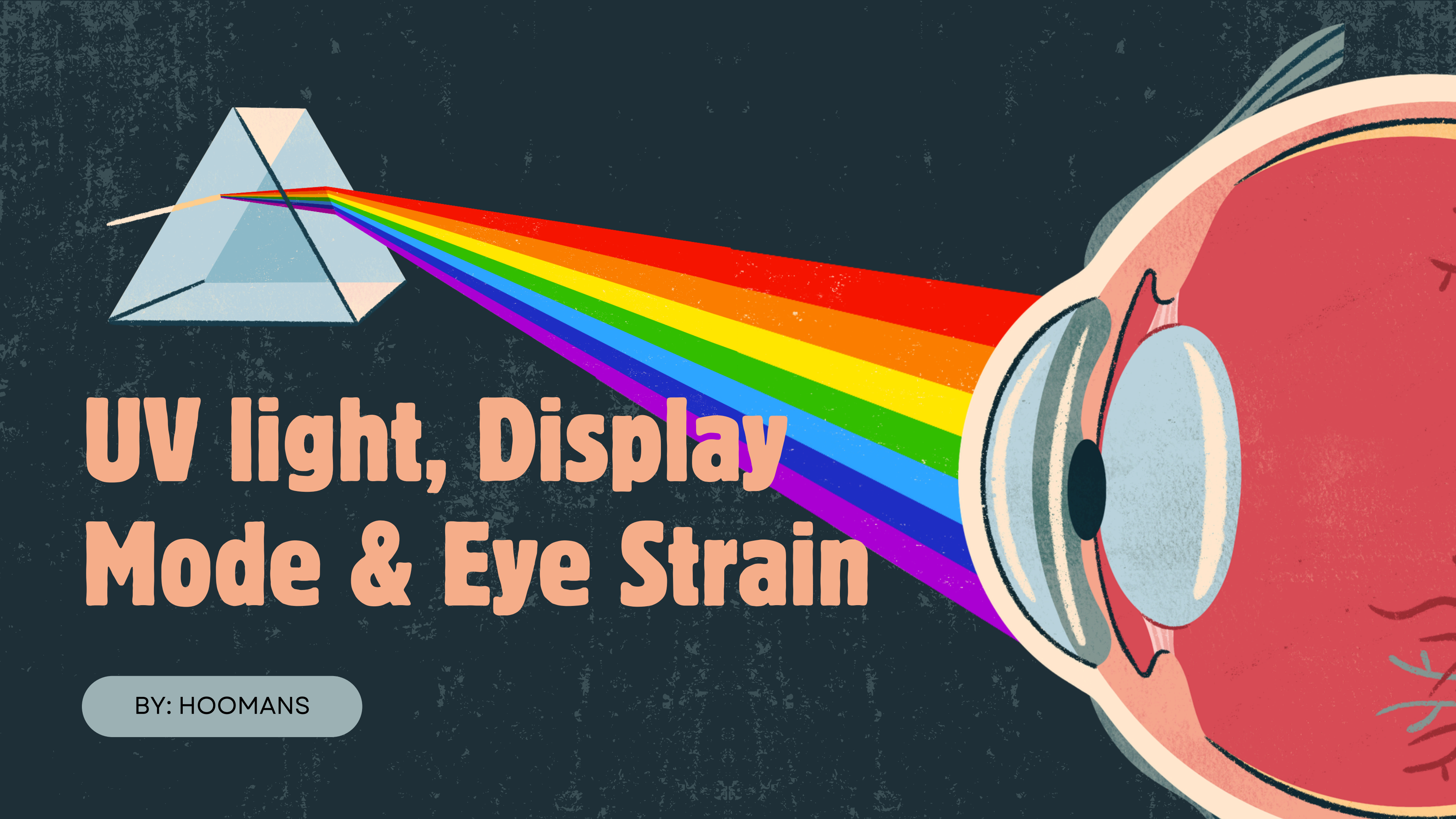




UV light, Display Mode & Eye Strain

BY: HOOMANS

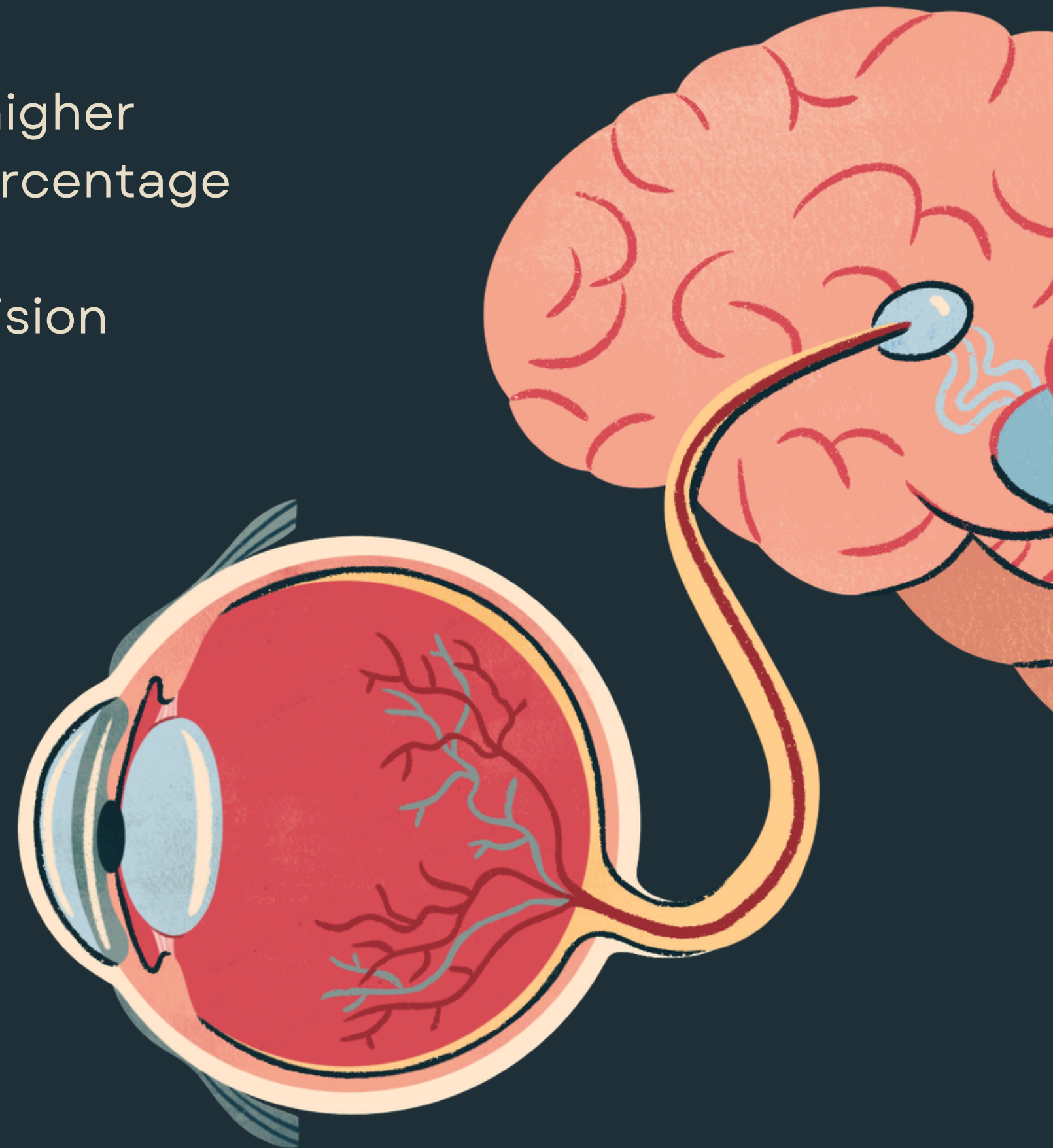
Why is it Important?

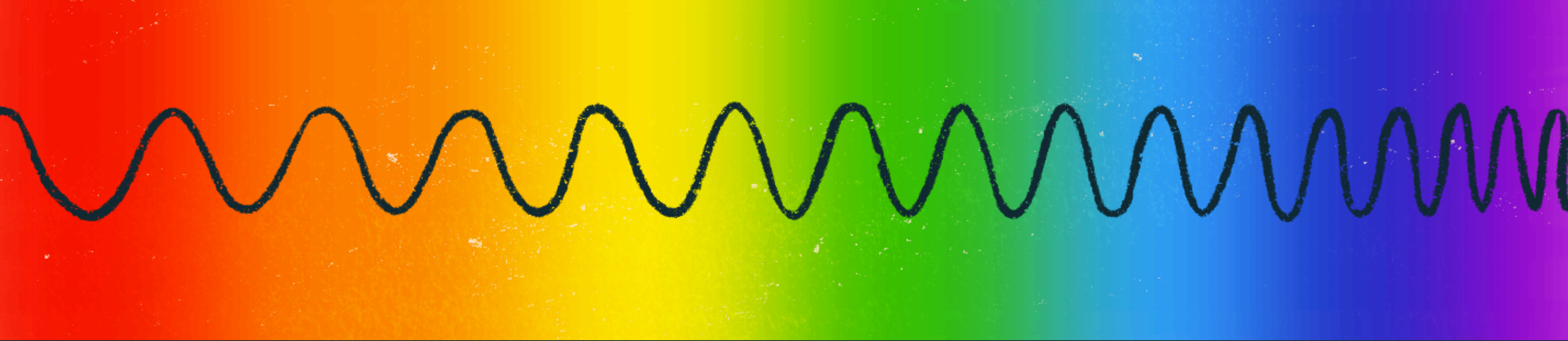
More usage of screens for academia especially for higher education has increased screen times for a huge percentage of students.

Common Issues: Eye fatigue, Dryness, and Blurred vision

Why this study matters?

Modern Education systems require students to have prolonged screen exposure, in result some false believes emerge amongst the users.....





Light and Wavelengths

Visible light is part of the electromagnetic spectrum. Wavelengths range from ~380 nm (violet) to ~750 nm (red). Blue light (400–500 nm) causes retinal stress & fatigue.

Color and Light

Objects absorb and reflect specific wavelengths.

The reflected wavelengths determine the color we perceive.

Research Gaps & Purpose:

Reasons:

- Most studies analyze variables separately
- Limited research on combined effects
- Study focuses on undergraduate students

Purpose:

- Effects of Blue-Light Filters
- Only for Under-graduate students
- No intervention



Research Question:

To what extent do blue light filtering and display mode (dark vs. light) independently and jointly affect self-reported eye strain among undergraduate students, when screen brightness, duration, and usage type are controlled?



Rod Cells



Cone Cells

Three Types of Cones

S-Cones (Short Wavelength)

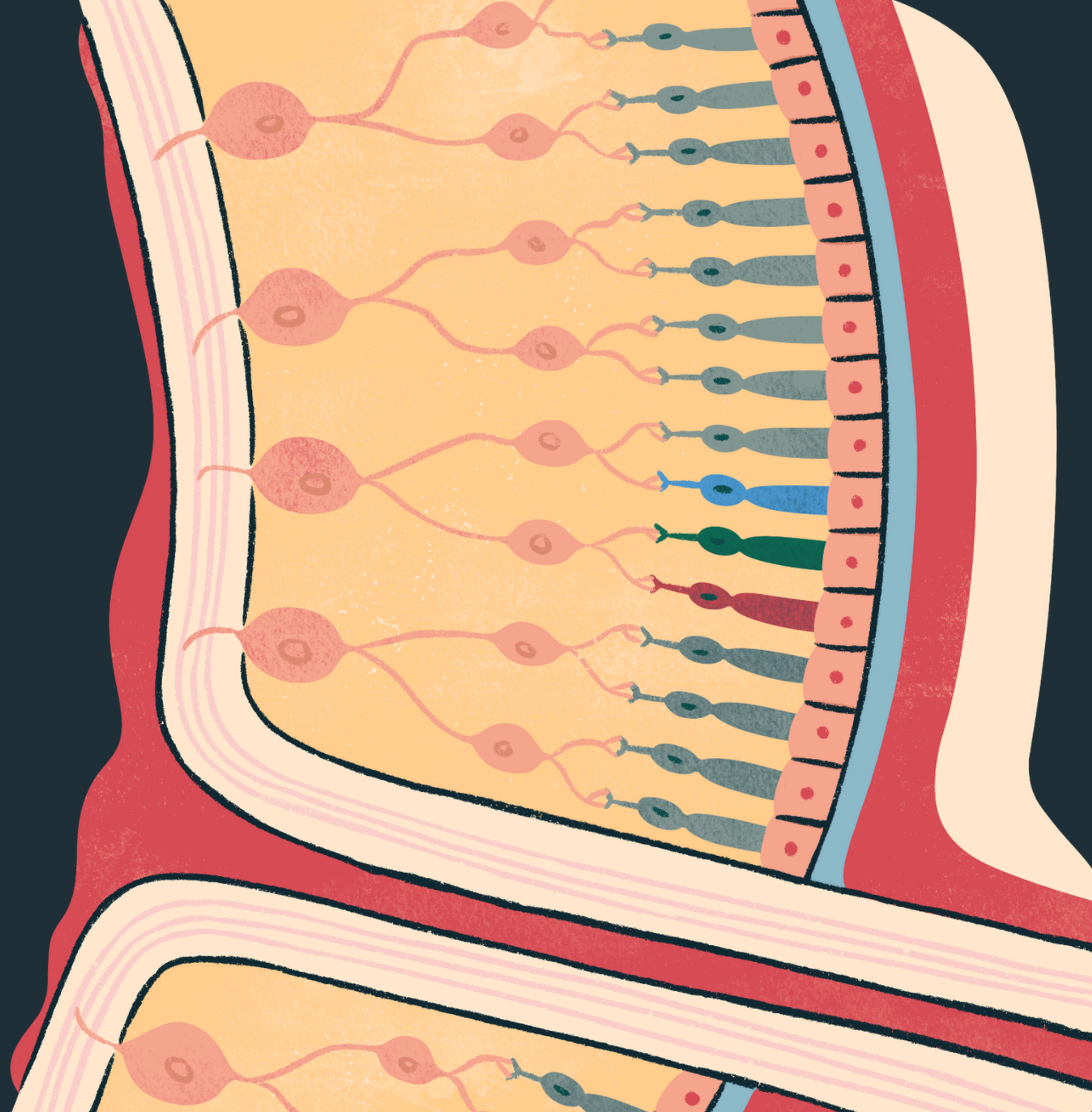
Sensitive to short wavelengths (blue).
Peak sensitivity around 420-440 nm.

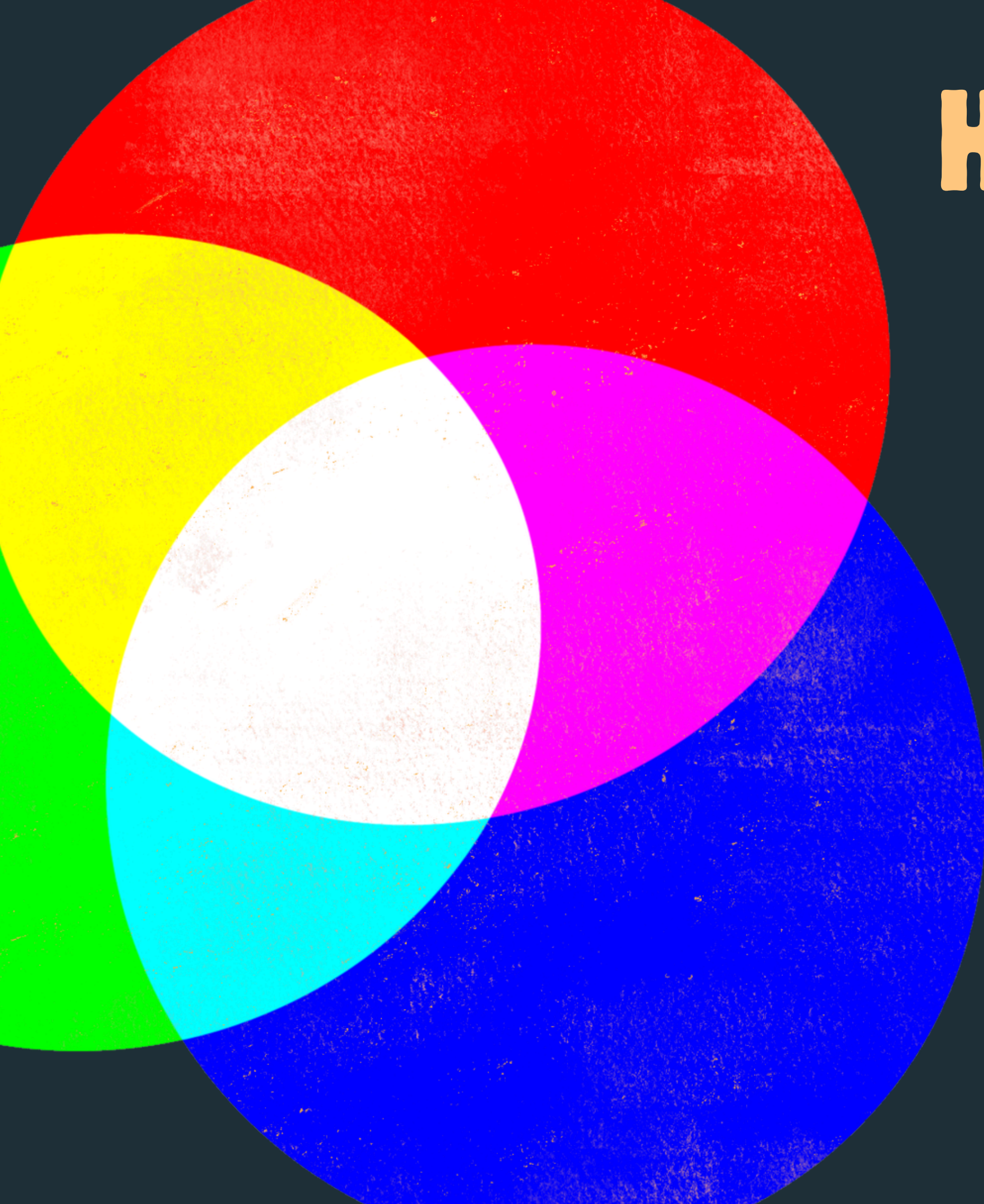
M-Cones (Medium Wavelength)

Sensitive to medium wavelengths (green).
Peak sensitivity around 530-540 nm.

L-Cones (Long Wavelength)

Sensitive to long wavelengths (red).
peak sensitivity around 560-580 nm





Hypothesis:

H1:

Blue light filter actually works and causes lower eye strain.

H2:

Dark mode helps in reducing eye strain.

Combined Prediction:

The lowest eye strain scores will be observed when both blue light filtering and dark mode are simultaneously active

H0:

No significant effect of Dark mode or Blue Light filters.

Research Design:

- Observational design
- Data collected at one point in time
- Identifies relationships, not causation

Participants / Sampling:

- Undergraduate Students
- Students of majors which require extensive time on screens.
- 100 - 150 samples
- Convenience sampling



Variables & Measurements



Dependent Variables:

- Eye Strain
- Self-reported
- On a scale from 1-10

Independent Variables:

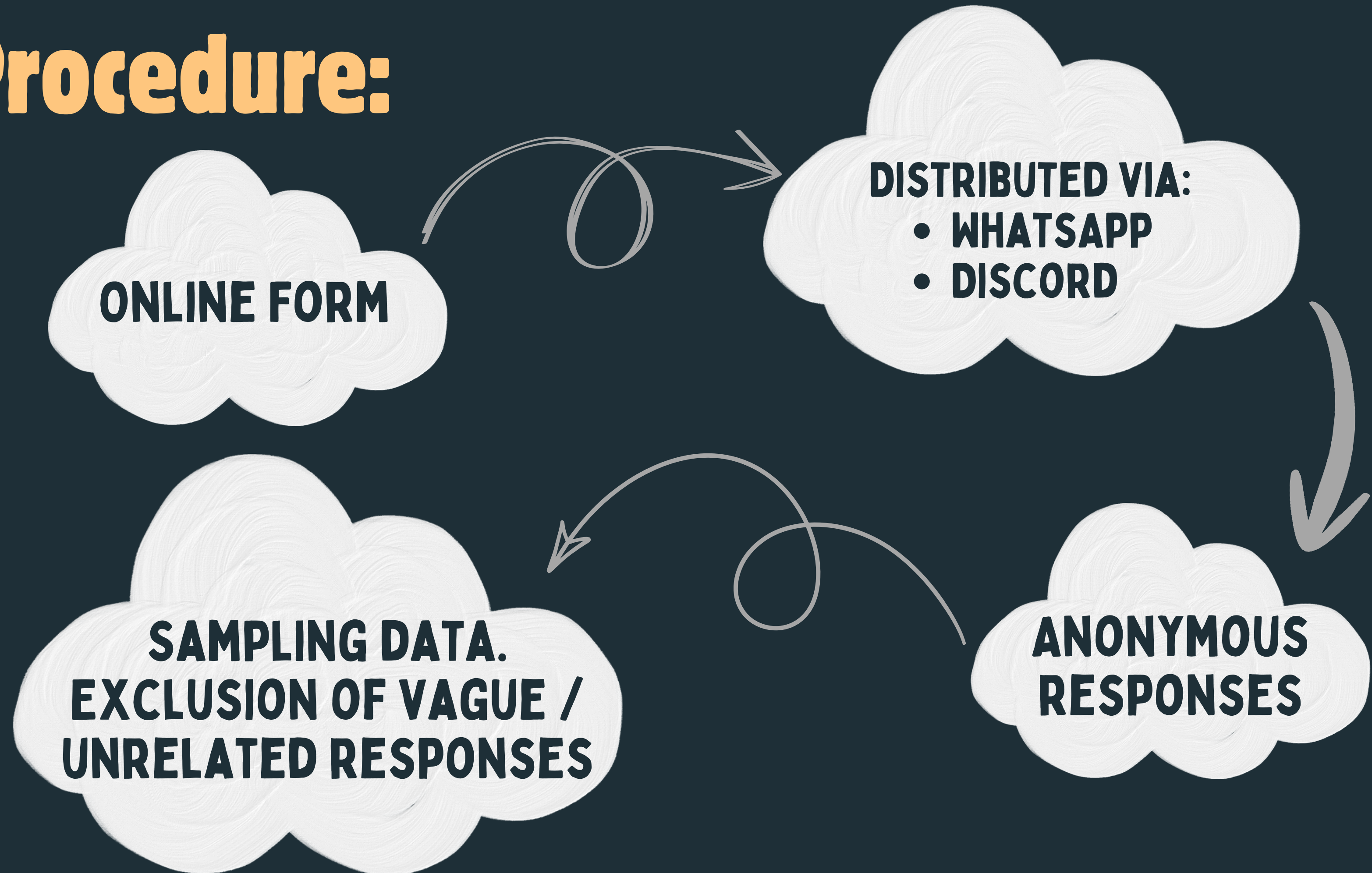
- Blue light filter (ON/OFF)
- Display mode (Dark/Light)

Controlled Variables:

- Screen time
- Device type
- Brightness
- Activity type



Procedure:



Data Analysis:

Descriptive statistics will summarize eye strain levels (mean, standard deviation, frequency distributions)

- Group comparisons will be conducted between:

Dark vs Light mode users

Blue light filter users vs non-users

Inferential analysis:

- Independent samples t-test for mean differences
- Linear regression to examine relationship with screen time
- Multiple regression to assess combined effects of variables





Expected Results:

Expected outcomes:

- Blue light filtering may reduce eye strain
- Dark mode may improve visual comfort
- Combined use may produce the lowest strain levels

If supported:

- Supports practical strategies for reducing eye strain

If not supported:

- Suggests other factors play a more significant role

Limitations:

- Self-reported data
- Observational design (no causation)

Future research:

- Experimental studies

Any Questions?



THANK YOU!!!